Juan David Munoz Bolanos

Junior Optical System Engineer

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Summary

Physicist and Optical Systems Engineer with extensive experience in the end to Polishing complex optomechatronic and medical devices from concept to series. Proven track record in translating physical concepts into robust prototypes using SolidWorks, structured V model systems engineering, and rigorous testing protocols. Adept at collaborating in interdisciplinary teams to deliver high quality technical documentation and system integration for clinical and industrial applications.

Research/Professional Experience

Jan 2022 to Jun 2026 — **Medical University of Innsbruck** — *Innsbruck, Austria*

Optical Systems Engineer | PhD Researcher

End to End System Design: Led the complete hardware to software build of a multiphoton microscope, integrating laser sources, galvo scanners, and adaptive optics. Achieved diffraction limited resolution.

Systems Engineering: Structured instrument development following V model principles, managing requirements tracking from concept to component testing. Utilized FMEA and 8D problem solving to prevent and resolve technical defects.

Quality & Testing: Executed structured Design of Experiments (DoE) and optical simulations in Python. Applied testing and documentation protocols for optomechanical assemblies with awareness of ISO 9001 and ISO 10110 frameworks.

CAD & Prototyping: Designed custom optomechanical mounts and alignment tools using SolidWorks and FreeCAD, ensuring precise mechanical tolerances and seamless integration.

Mar 2020 to Dec 2021 — **Leibniz Institute of Photonic Technology** — *Jena, Germany*

R&D Research Assistant

Spectrometer Prototyping: Designed and built a dispersive 1064 nm fiber probe Raman spectrometer for non invasive human tissue characterization, handling optical layout and detector calibration.

Data Analysis & ML: Developed a custom Python package (SpectraMap) to apply supervised machine learning algorithms to 2D hyperspectral Raman images, achieving high accuracy classification of tissue samples.

Technical Documentation: Authored comprehensive technical reports, system manuals, and peer reviewed publications detailing optical design parameters and calibration procedures.

Technical Competencies

Optics & Hardware: Adaptive optics · Multiphoton/confocal microscopy · Raman spectroscopy · Laser systems (fs, CW, 1064 nm) · Camera integration · 3D CAD (SolidWorks, FreeCAD)

Software & Tools: Python (JAX, scikit-learn, NumPy) · LabVIEW/FPGA · C++ · OpenCV · Git · MATLAB

Soft Skills: Interdisciplinary teamwork · Fast learner on new toolchains

Education

Jan 2022 to Jun 2026 — **Medical University of Innsbruck** — *Innsbruck, Austria*
Doctor of Philosophy in Optical Metrology

Oct 2019 to Sep 2021 — **Friedrich Schiller Univ. Jena** — *Jena, Germany*
Master of Science in Photonics

Aug 2012 to Aug 2018 — **University of Cauca** — *Popayán, Colombia*
Bachelor of Physics Engineering

Publications

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Systems & Quality Engineering: V-Model, 8D Problem Solving, Design of Experiments, ISO 9001.

Adaptability & Independent Working: Self-driven approach to bringing concepts to working systems; comfortable in dynamic, fast-paced environments.

Teamwork & Communication: Cross-functional collaboration across optics, hardware, software, and production teams; structured root cause analysis.

Personal Interests

Biking/Hiking | Bachata Dancing | Harmonica | Learning German

Languages

English (Fluent) | German (Intermediate B2) | Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte
Medical University of Innsbruck
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PD Dr. Christoph Krafft
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